Indian Institute of Technology Bombay

PH 544 - Quiz 2 | **General relativity** | **Date:9-04-2026**
Marks 10 | **LC-002** | **Time: 17:30-18:30**

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1. Einstein's GR alters the solution of the central force problem in Newtonian gravity. The relative coordinate measured from the center of mass (CM) becomes
$$\frac{1}{r} = \frac{GM}{l^2}\left[1 + e(\cos(\phi - \epsilon\phi))\right]$$
where the perturbation parameter $\epsilon \equiv \left(\frac{3GM}{lc}\right)^2$, $l = \sqrt{GMa(1-e^2)}$ is angular momentum per unit mass, M is the total mass and ϕ is the polar coordinate in the orbital plane with $\phi = 0$ corresponds to the closest point to the CM. Using this, Mercury's orbit with 88 day period, orbital eccentricity of 0.21 and perihelion advance of 43 arcsecs per century can be explianed. If Pluto's orbital period is 248 years and orbital eccentricity of 0.25, can you estimate the perihelion advance of Pluto's orbit in its one orbital period? **[2]**

2. Consider the line element of the static space-time as
$$dl^2 = c^2dt^2 - a^2\left(\frac{d\sigma^2}{1-k\sigma^2} + \sigma^2 d\theta^2 + \sigma^2\sin^2\theta d\phi^2\right)$$
where σ is the dimensionless radial co-ordinate with maximum value 1 and a is the cosmic scale factor. In general, k can have values $0, \pm 1$. Compute the volume of the three sphere in this universe. **[3]**

3. Astronauts wish to orbit the supermassive black-hole at the center of our galaxy. Assume the metric is:
$$ds^2 = \left[1 - \frac{2GM}{c^2 r}\right]dt^2 - \frac{dr^2}{[1 - 2GM/(c^2 r)]} - r^2\left(d\theta^2 + \sin^2\theta d\phi^2\right)$$
They follow a circular orbit with $r = R, \theta = \pi/2$ and constant angular velocity $d\phi/dt = \Omega$.

   (a) Calculate the four-acceleration, $a^\mu = u^\nu \nabla_\nu a^\mu$ and its magnitude $a = \sqrt{a^\mu a_\mu}$ as a function of $R, \Omega,$ and M.
   Here, $\nabla_\alpha A^\beta = \partial_\alpha A^\beta + \Gamma^\beta_{\alpha\delta}A^\delta$. **[4]**

   (b) Obtain Ω for this circular orbit. Is this orbit freely falling? Please justify. **[1]**